

MappingType#

<https://pytorch.org/docs/stable/generated/torch.quantization.observer.Map>

Description:

How floating point number is mapped to integer number symmetric mapping means floating point range is symmetrically mapped to integer range let's say we have floating point range $(-3.5, 10.2)$ and integer range $(-8, 7)$ (int4) we'll use $(-10.2, 10.2)$ as the range for floating point and map that to $(-8, 7)$ e.g. $\text{scale} = (10.2 - (-10.2)) / (7 - (-8))$

SYMMETRIC_NO_CLIPPING_ERR is a variant of symmetric mapping, where the scale is the max of smin and smax , where $\text{smin} = \text{min_val_neg} / \text{quant_min}$, and $\text{smax} = \text{max_val_pos} / \text{quant_max}$. By calculating smin and smax individually, there can be less round error on negative values, and no out-of-range of all floating point values. asymmetric mapping means we just directly map the floating point range to integer range, for the above example, we will map $(-3.5, 10.2)$ to $(-8, 7)$ and calculate quantization parameter based on this mapping e.g. $\text{scale} = (10.2 - (-3.5)) / (7 - (-8))$

Function Signature

```
class torch.ao.quantization.observer.MappingType(value)  
[source]#
```

Detailed Documentation

Documentation

How floating point number is mapped to integer number

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